



University of Wisconsin
**SCHOOL OF MEDICINE
AND PUBLIC HEALTH**

Causality as Minimum Energy Principle

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EMBC July 27, 2026 2:30pm

Abstract: Human brain networks exhibit dense recurrent interactions that are poorly represented by directed acyclic graph (DAG) assumptions or edgewise analyses alone. We introduce a variational framework for causal inference without acyclicity, where directed neural interactions are modeled as flows on a simplicial complex and evolved under an energy-minimizing dynamical system. This separates transient components from persistent harmonic flows, which define a low-dimensional Hilbert space of stable directed cycles. By treating cycles as primary objects of inference, the framework allows recurrent organization to be projected, compared, and statistically analyzed as a global geometric object rather than as isolated edges, paths, or local motifs. Applied to resting-state fMRI, the method reveals densely interacting but stable cyclic structures that reconfigure over time while remaining within the same cyclic subspace. These results suggest that dense cyclic interaction is a fundamental organizing principle of human brain networks and provide a scalable vector-space framework for statistical inference on recurrent dynamics in high-dimensional systems. The talk is based on [arXiv:2604.17151](https://arxiv.org/abs/2604.17151).

Talk is based on

CAUSALITY AS A MINIMUM ENERGY PRINCIPLE

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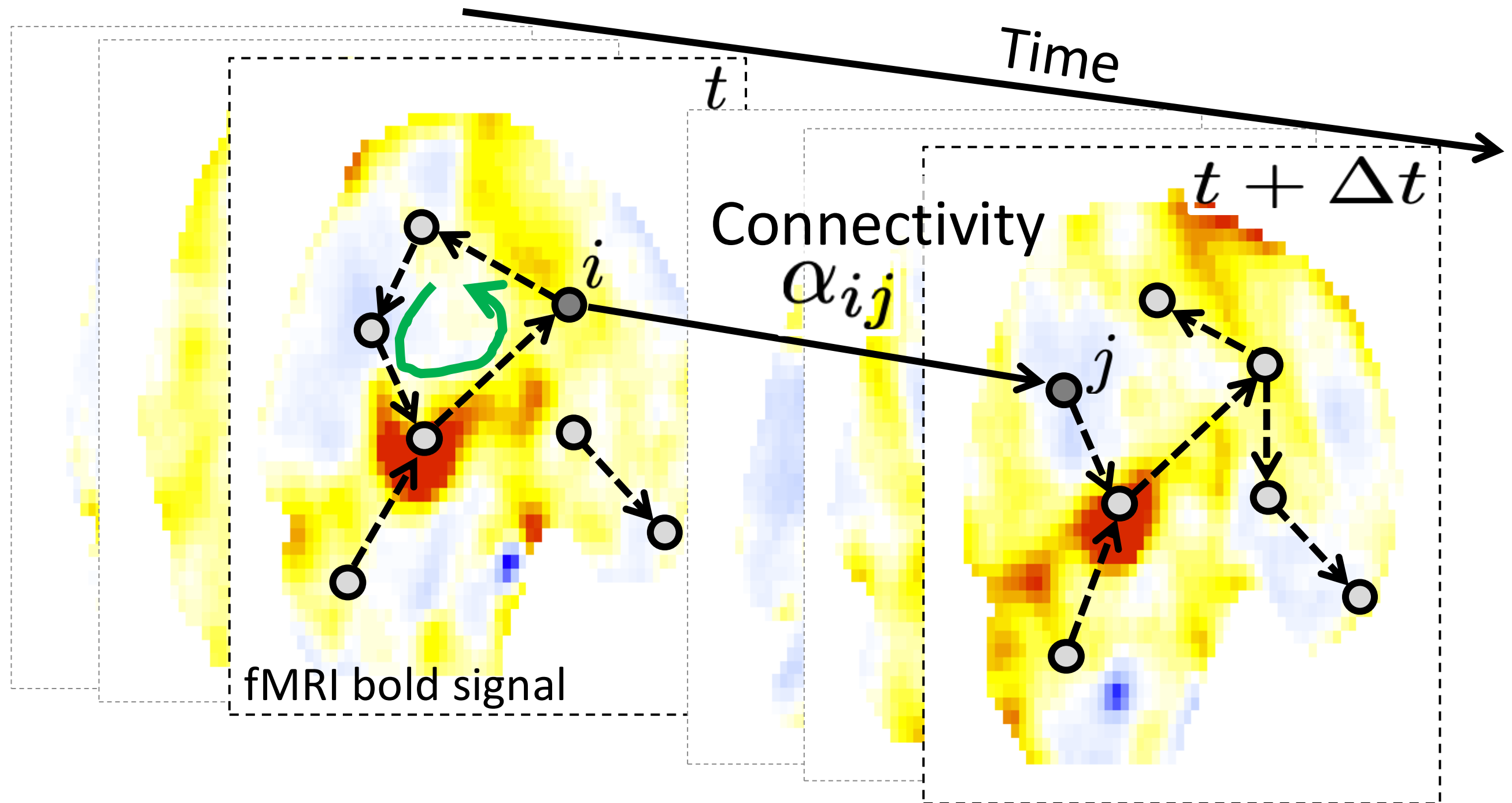
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[arXiv:2604.17151](https://arxiv.org/abs/2604.17151)

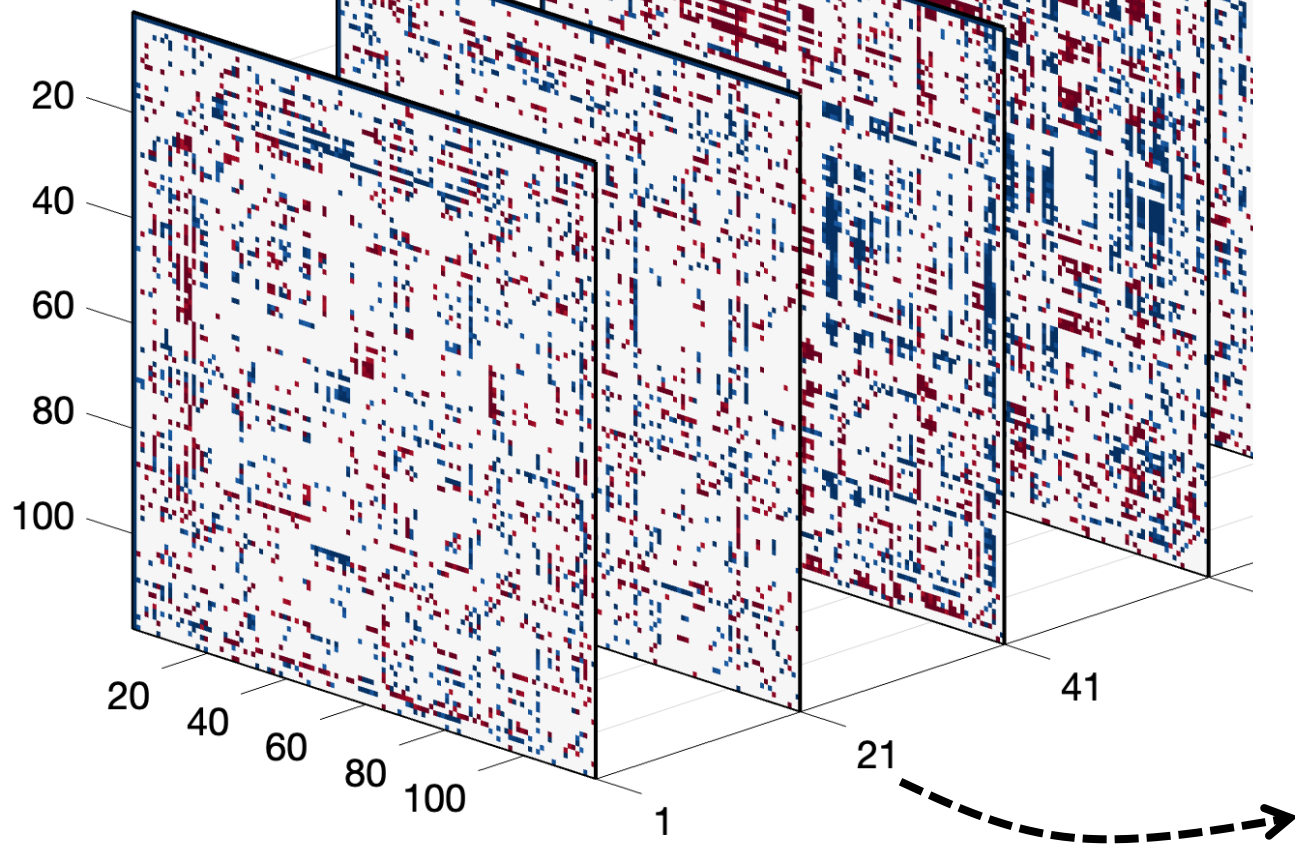
Traditional Causal Models for rs-fMRI

Granger Causality, Structural Equation Model (SEM), Dynamic Causal Model (DCM)

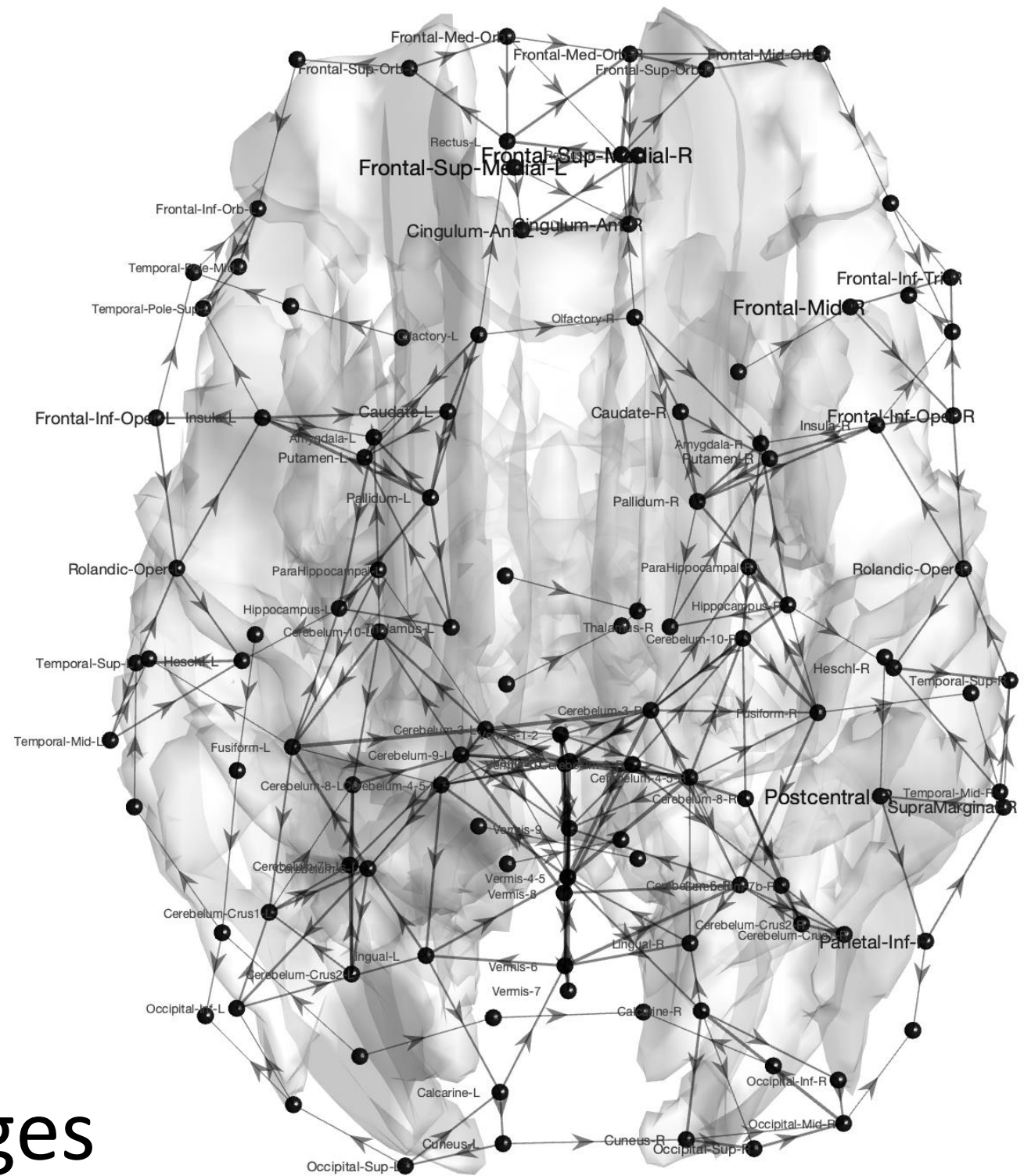


$$x_j(t + \Delta t) = \sum_i \alpha_{ij} x_i(t) + \cdots + e_j$$

Input data HCP rs-fMRI



Traditional causal models
only encode pairwise
interactions as directed edges



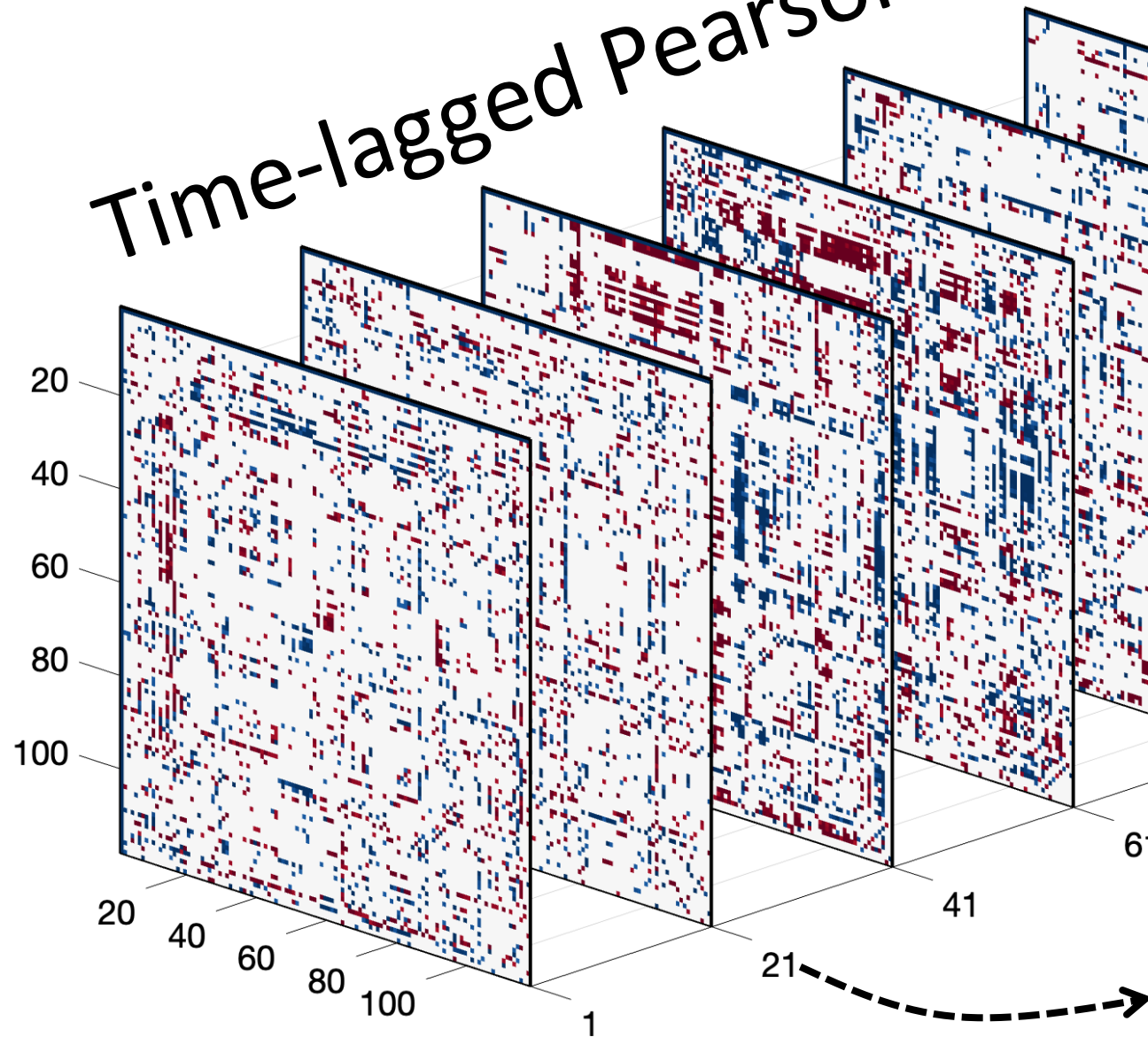
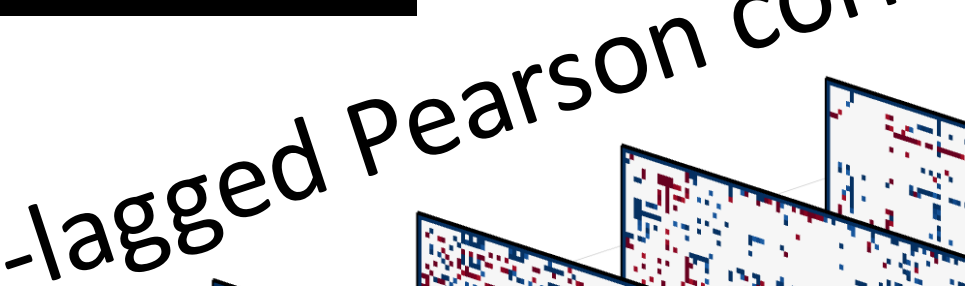
Out approach:
Model agnostic

Time-lagged Pearson correlation

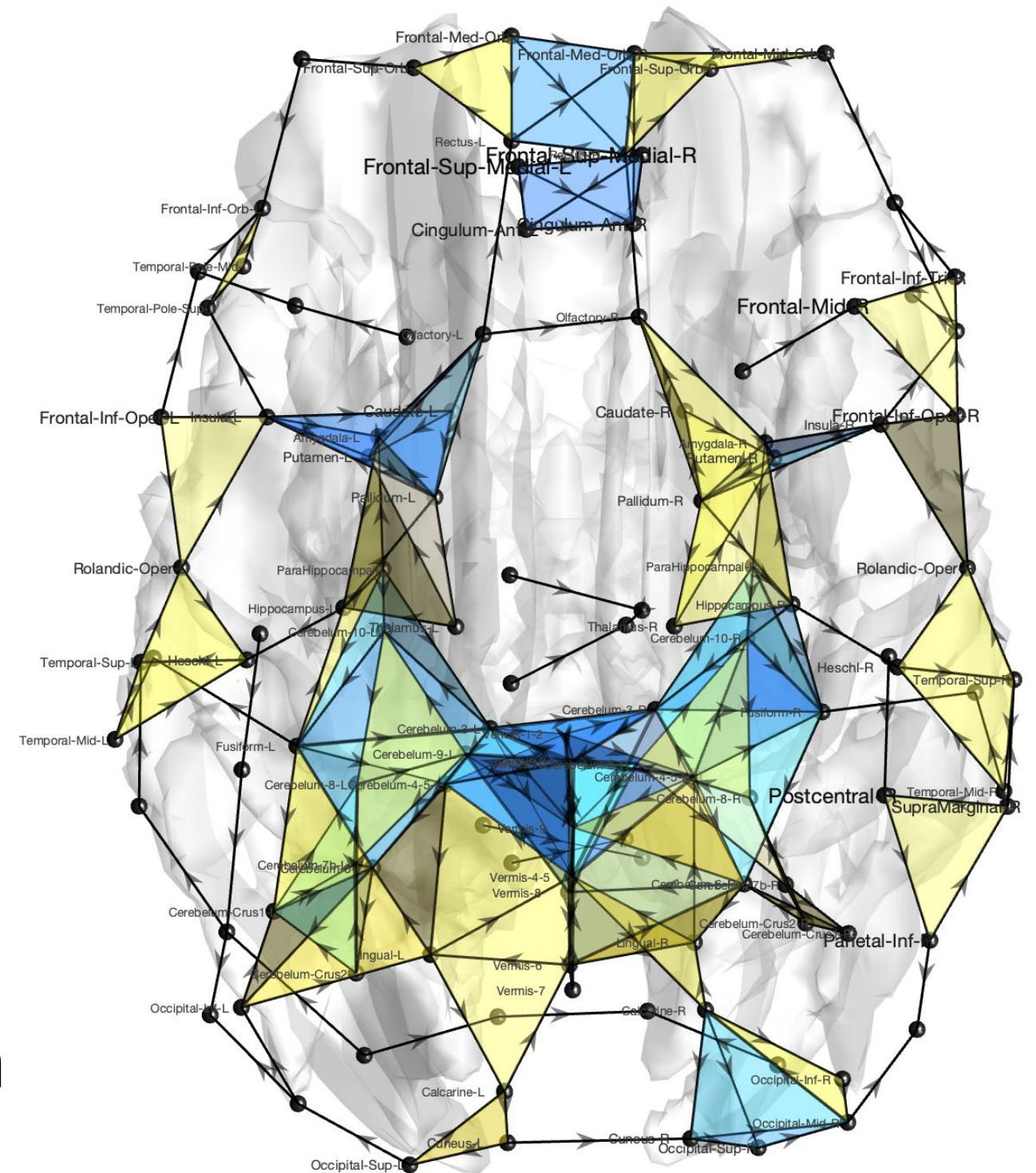
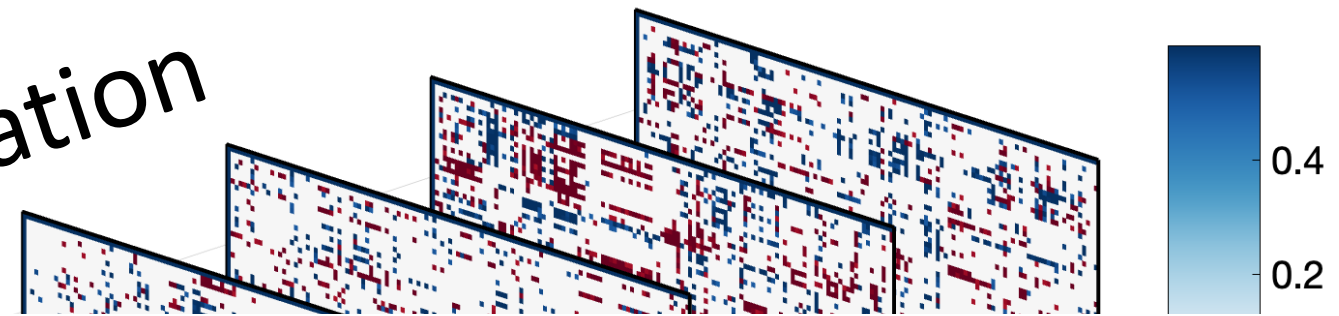


model agnostic

Time-lagged Pearson correlation



Simplicial complex encode
higher-order interactions as an
element in a **vector space**

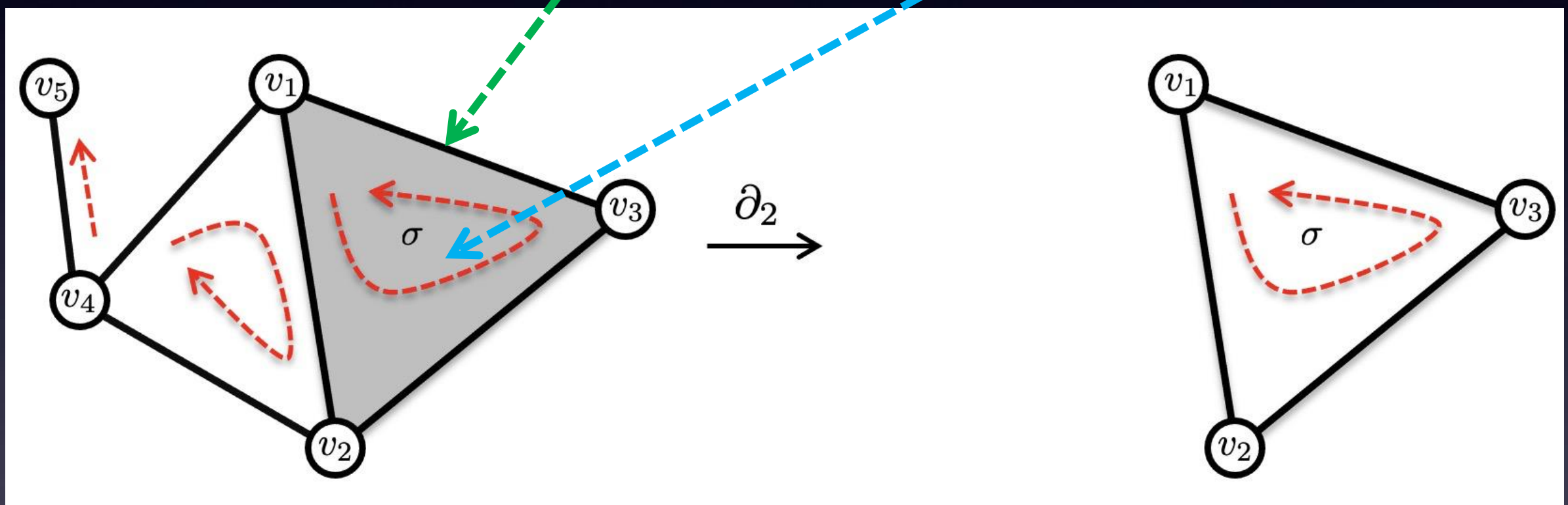


1st Hodge Laplacian

*Anand and Chung, 2023, IEEE
TMI*

Anand and Chung, 2024 ISBI

$$\Delta_1 = \overset{\substack{\text{\# of nodes} \\ \text{\# of edges}}}{\mathbf{B}_1^T \mathbf{B}_1} + \overset{\substack{\text{\# of edges} \\ \text{\# of faces}}}{\mathbf{B}_2 \mathbf{B}_2^T}$$



0th Hodge Laplacian
Graph Laplacian

$$\Delta_0 = \overset{\substack{\text{\# of edges} \\ \text{\# of nodes}}}{\mathbf{B}_1^T \mathbf{B}_1}$$

Causality as Minimum Energy Principle

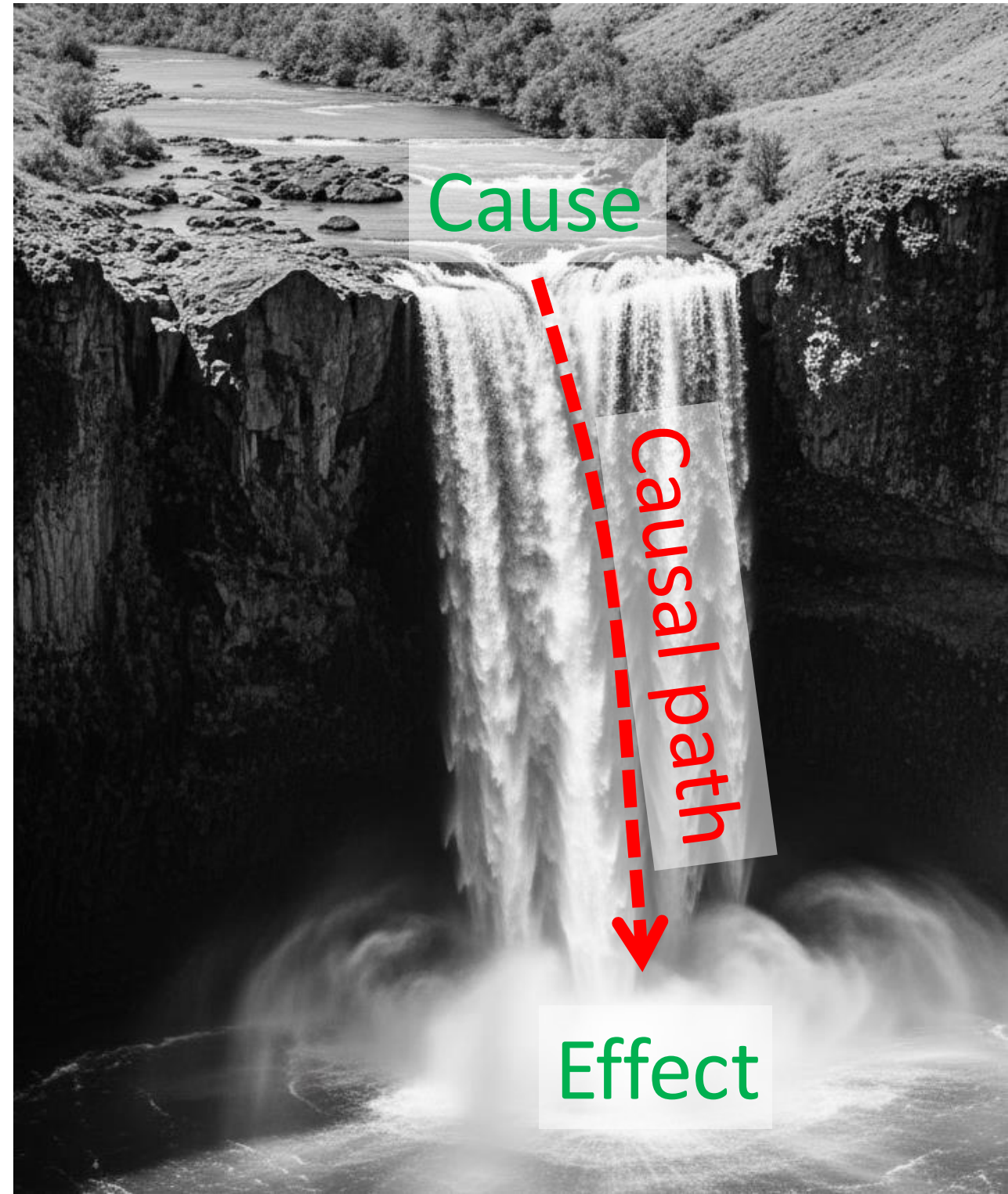
$$\begin{aligned} E(X) &= \frac{1}{2} \langle X, \mathcal{L}_1 X \rangle && \text{Dirichlet energy} \\ &= \frac{1}{2} \underbrace{\|\mathbf{B}_1 X\|_2^2}_{\text{divergence}} + \frac{1}{2} \underbrace{\|\mathbf{B}_2^\top X\|_2^2}_{\text{curl}} \end{aligned}$$

*The steepest
gradient of energy*

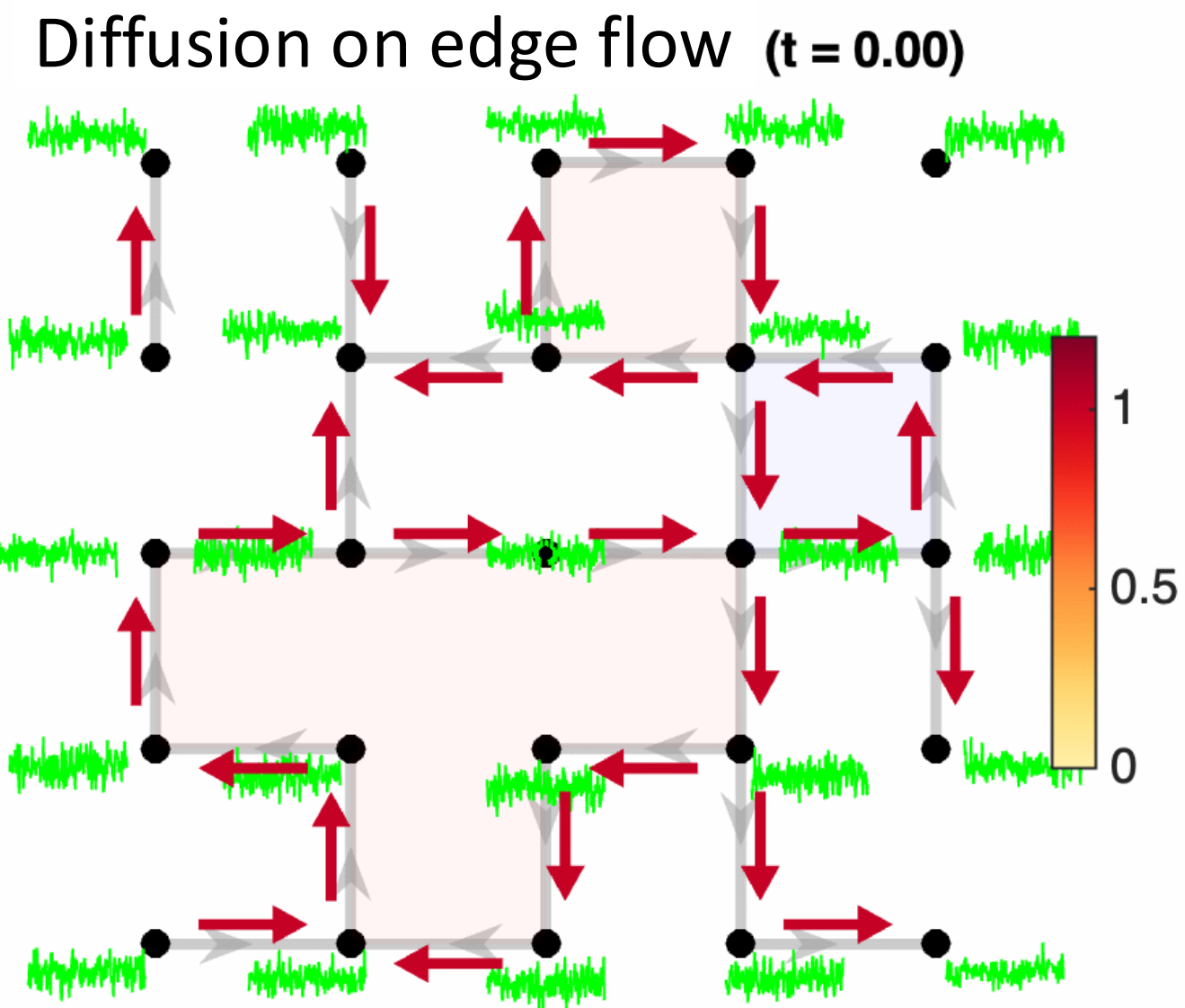
$$\frac{dX(t)}{dt} = -\nabla E(X(t))$$

Causal propagation

$$\frac{dX(t)}{dt} = -\Delta_1 X(t)$$



Dirichlet-Hodge diffusion



$$\dot{X}(t) = -\Delta_1 X(t)$$

$$X(t) = e^{-\Delta_1 t} X(0)$$

Harmonic flow

$$X_H = \mathcal{P}_H X(0)$$

Orthogonal
projection

$$\sum_k \psi_k \psi_k^\top$$

Codes

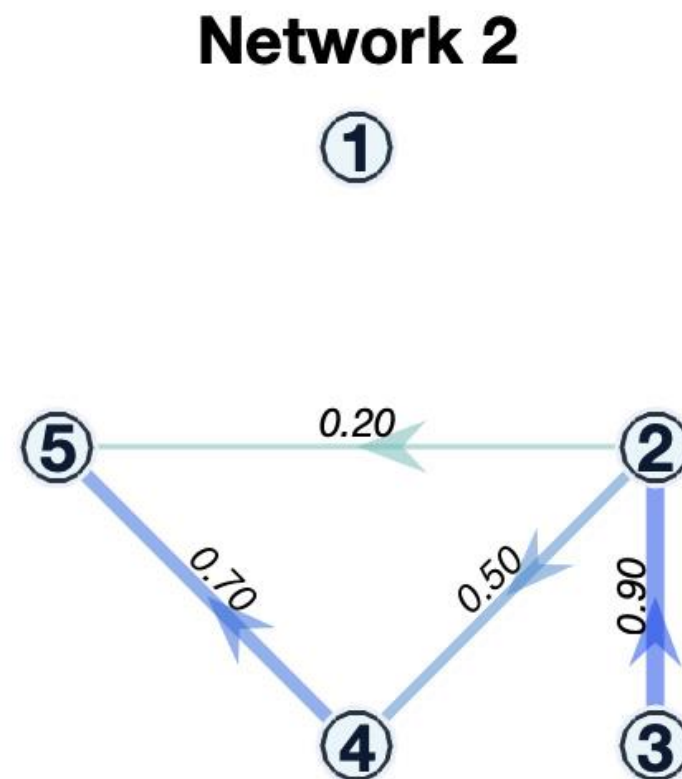
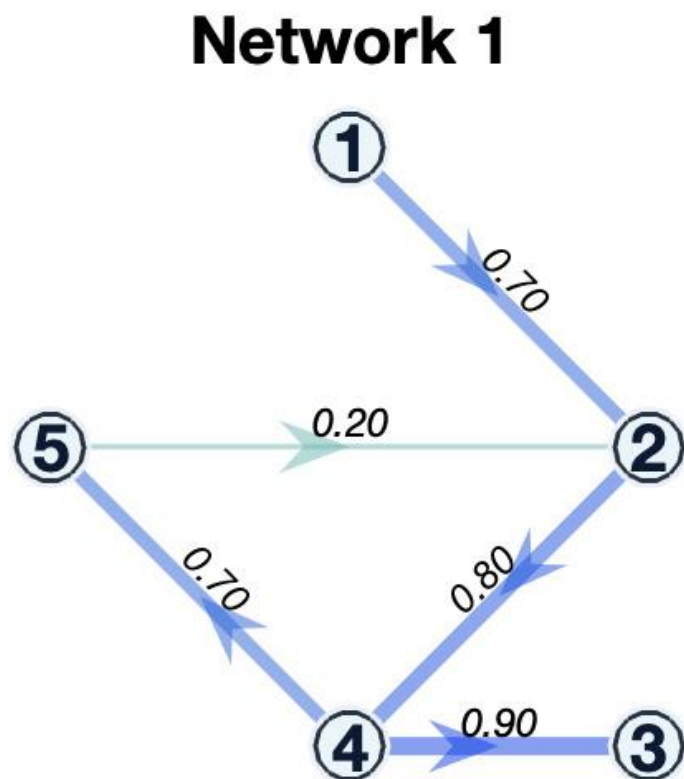
[https://github.com/
laplacebeltrami/harmoni](https://github.com/laplacebeltrami/harmoni)

$\ker(\Delta_1)$ Hilbert space of cycles

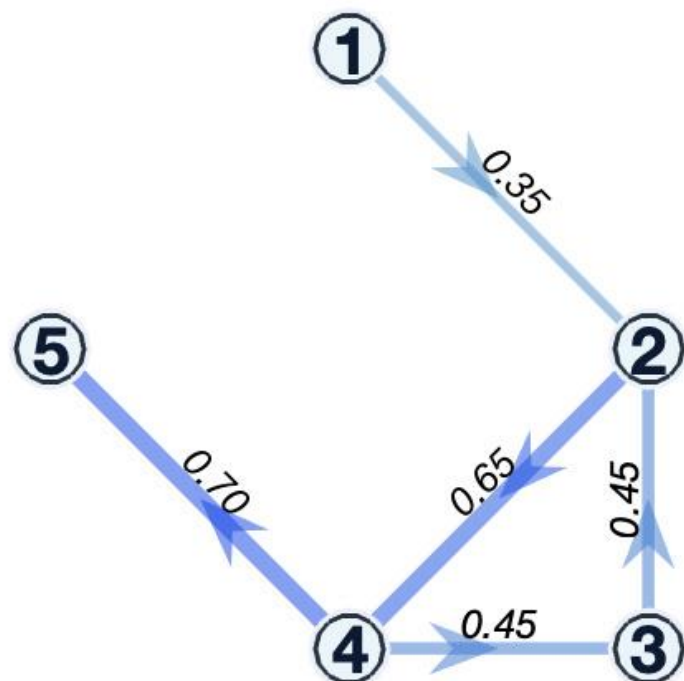
Cycles

$$C_1, C_2 \in \ker(\Delta_1)$$

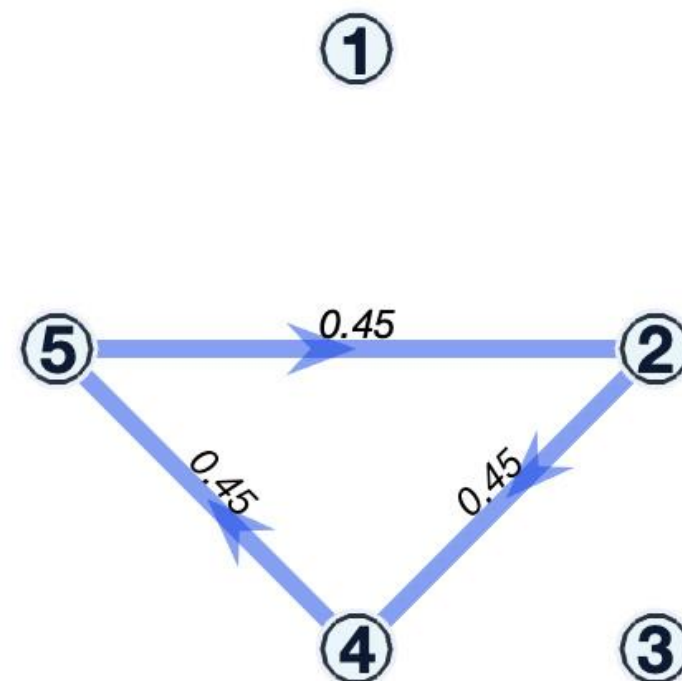
$$\alpha_1 C_1 + \alpha_2 C_2 \in \ker(\Delta_1)$$



Algebraic Averaging



Averaging in Cycle Space



Inner product

$$\langle C_1, C_2 \rangle = C_1^\top C_2$$

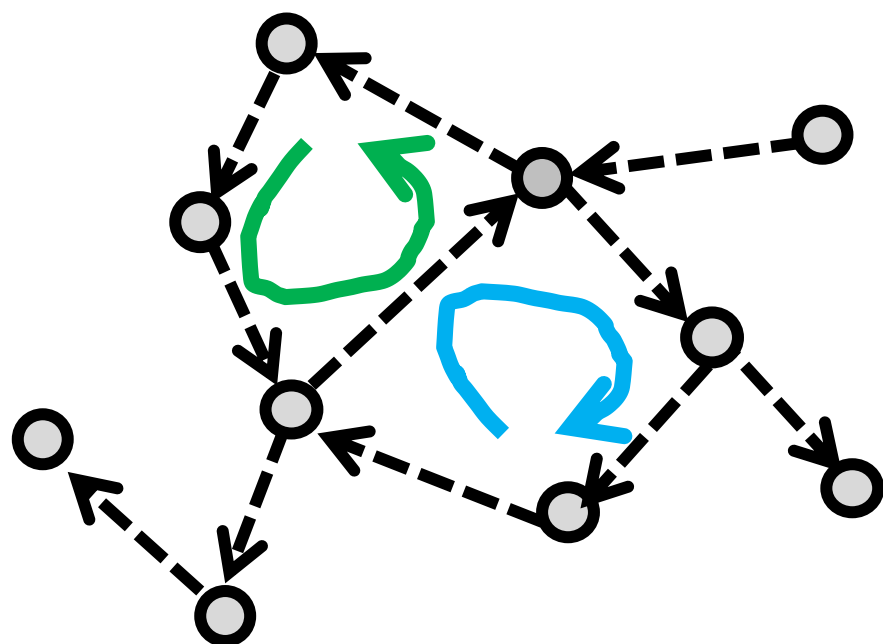
Stability of harmonic flow

Theorem (Variance reduction of harmonic flow). *Let $X \in \mathbb{R}^{|E|}$ be a random edge flow with isotropic covariance $\Sigma_X = \sigma^2 I$. Then the total variance of the harmonic flow $X_H = \mathcal{P}_H X$ satisfies*

$$\text{Var}(X) := \mathbb{E} \|X - \mathbb{E}X\|_2^2$$

$$\text{Var}(X_H) = \frac{\beta_1}{|E|} \text{Var}(X),$$

where $\beta_1 = \dim \ker(\Delta_1)$ is the first Betti number, equal to the dimension of the harmonic subspace.



Sparse

$$\beta_1 = 2$$

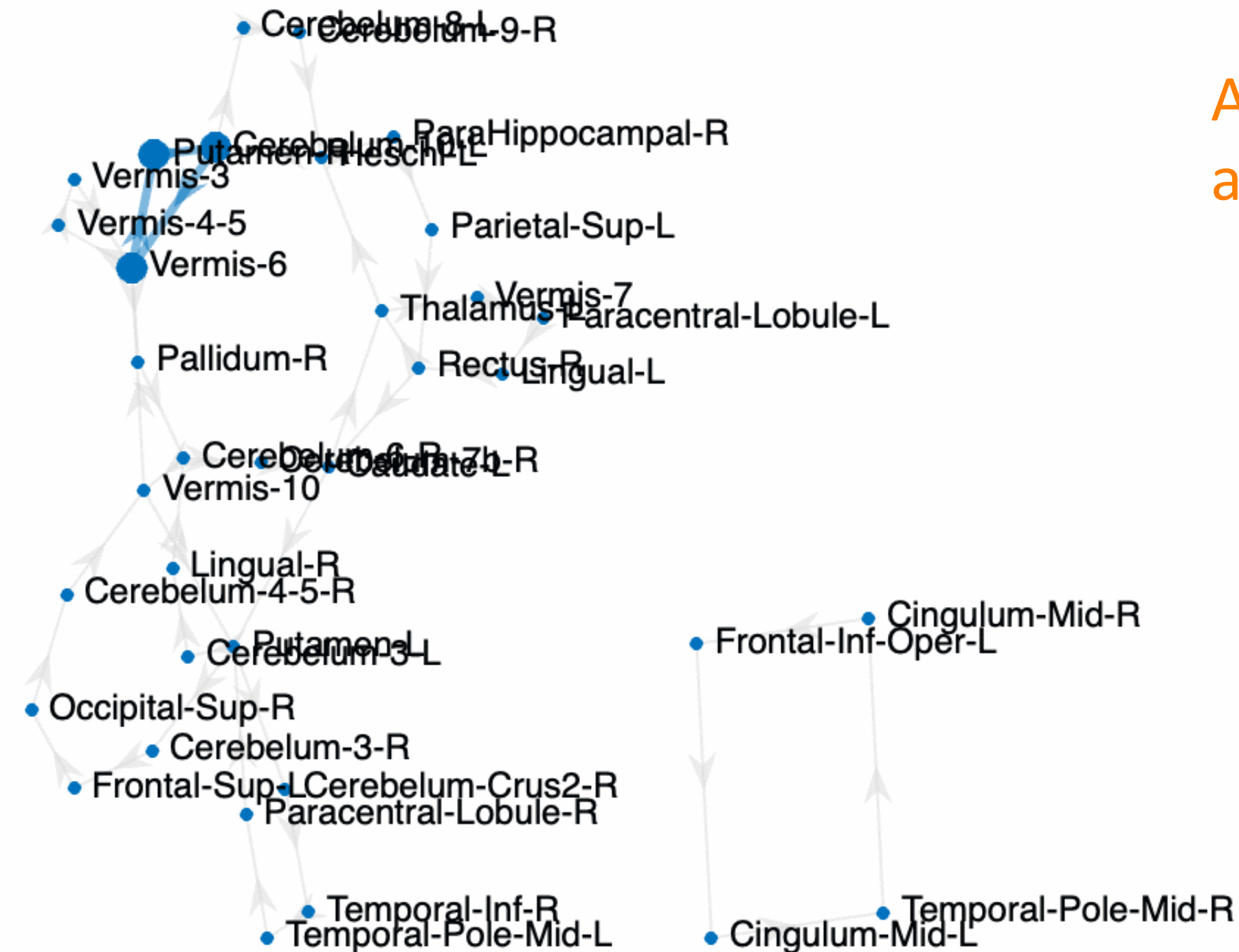
$$|E| = 11$$

Complete

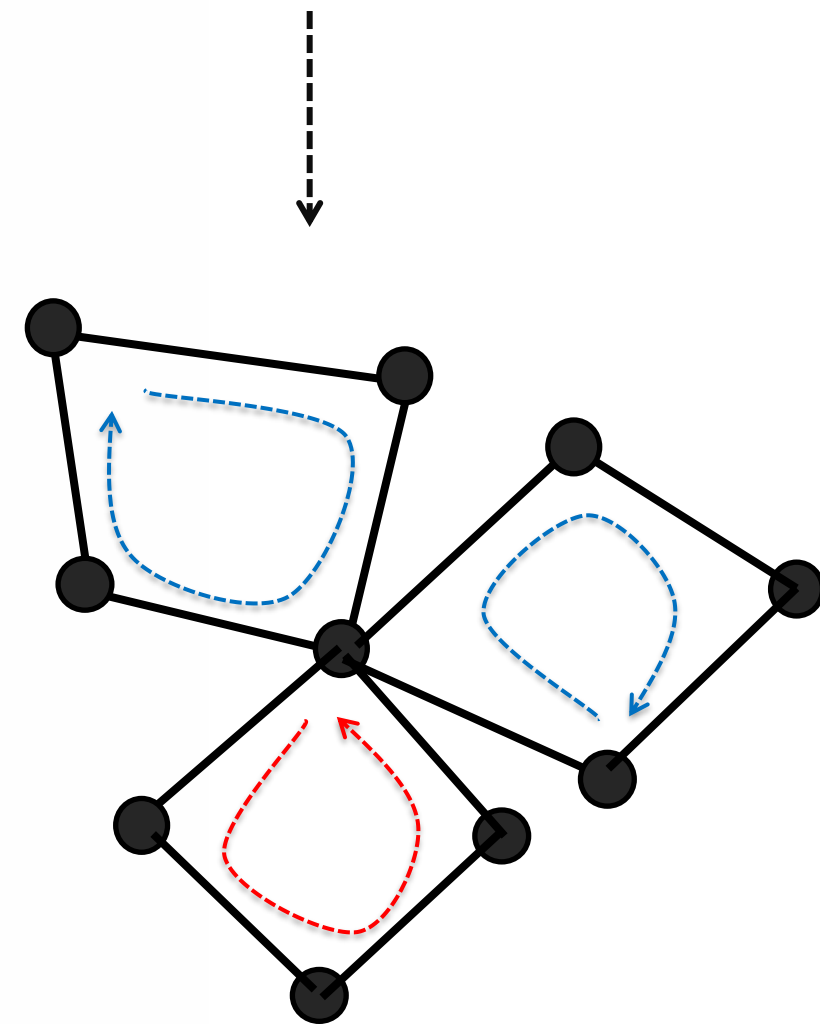
$$\frac{\beta_1}{|E|} = 1 - \frac{2}{|V|}$$

Cyclic interactions in rs-fMRI network

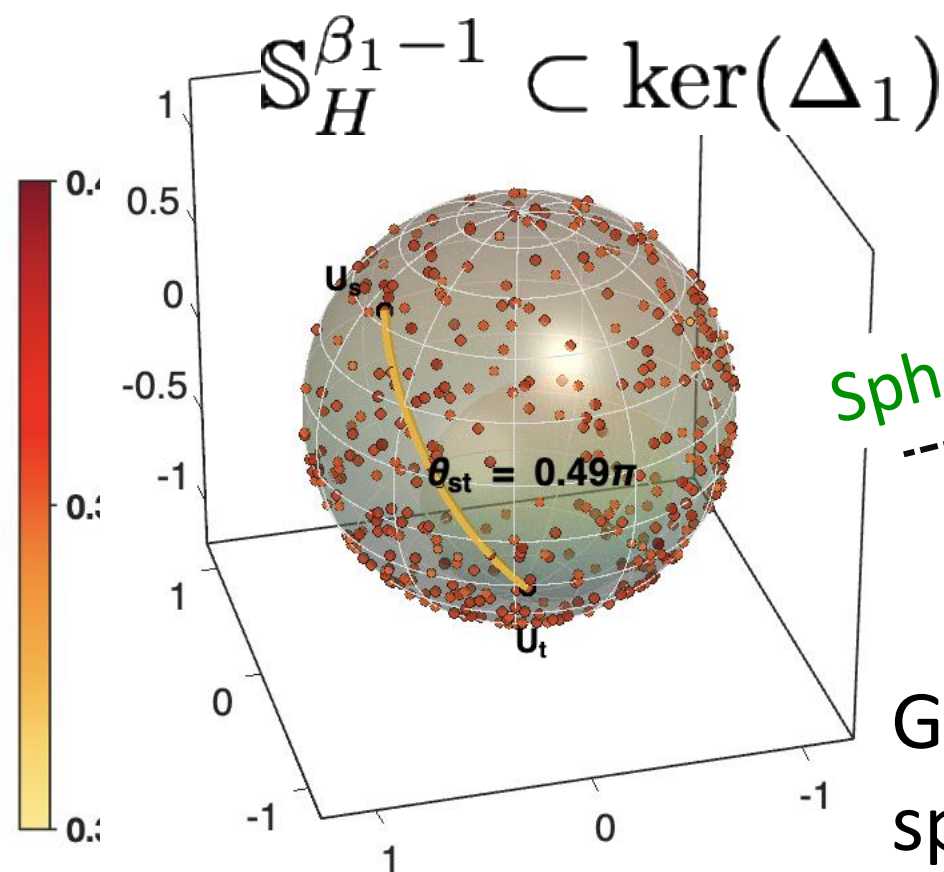
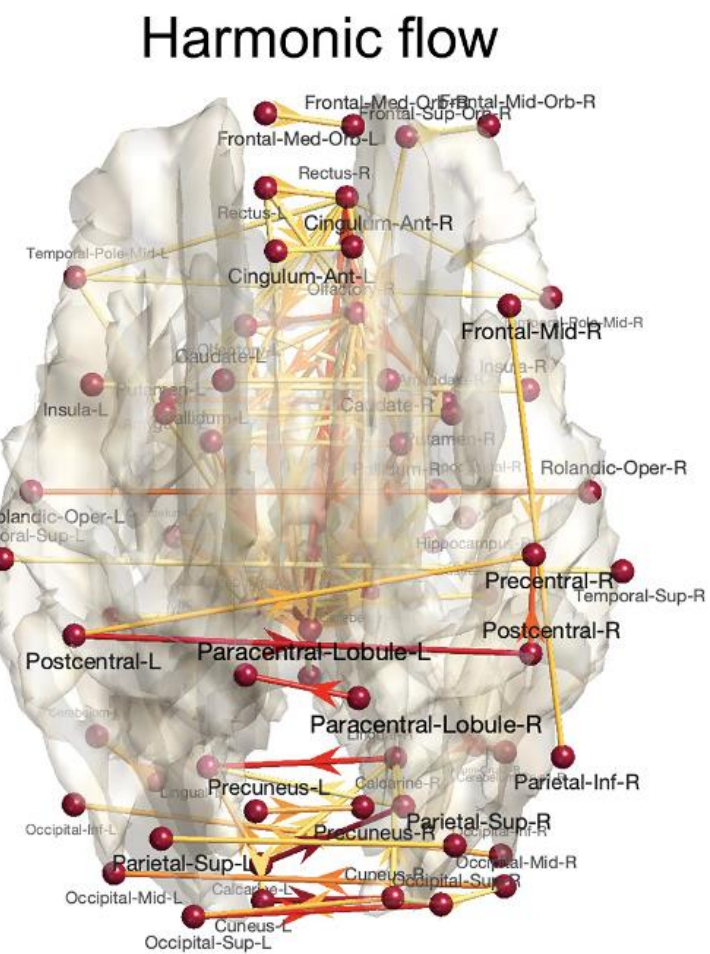
Cycles 1..1 $\kappa=0.3017$



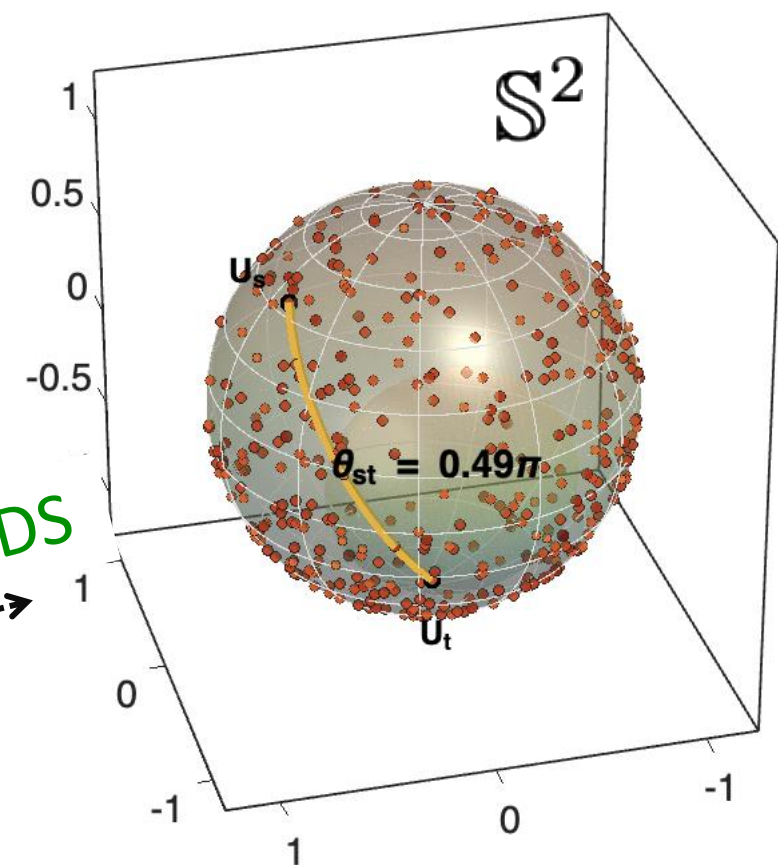
All 534 harmonic cycles
are node-shared



Inference in cyclic space



Spherical-MDS



Geodesics are invariant across spheres of all dimensions.

Flows are mapped to a point in harmonic sphere

$$U^{(s)} = \frac{X_H^{(s)}}{\|X_H^{(s)}\|}$$

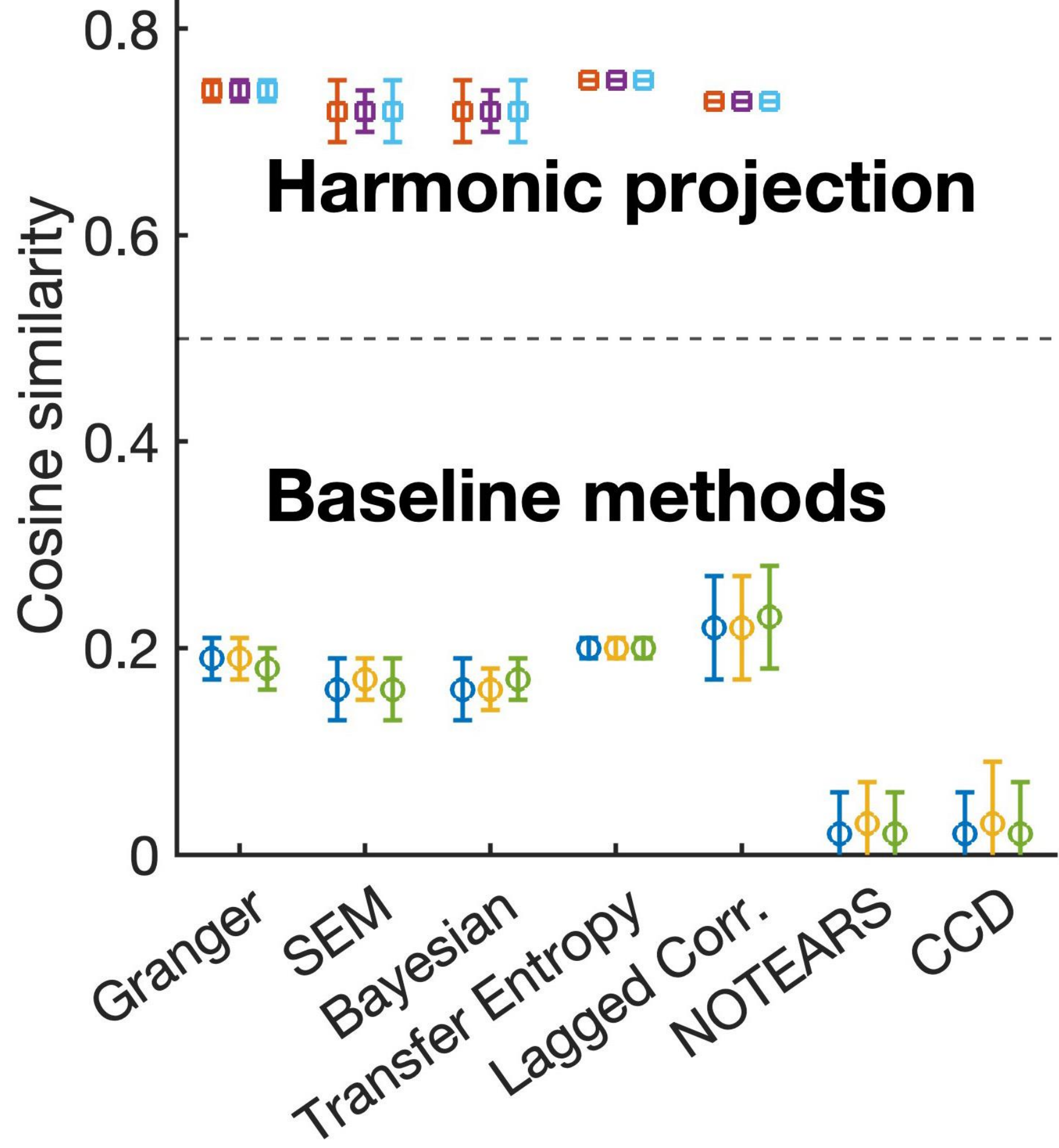
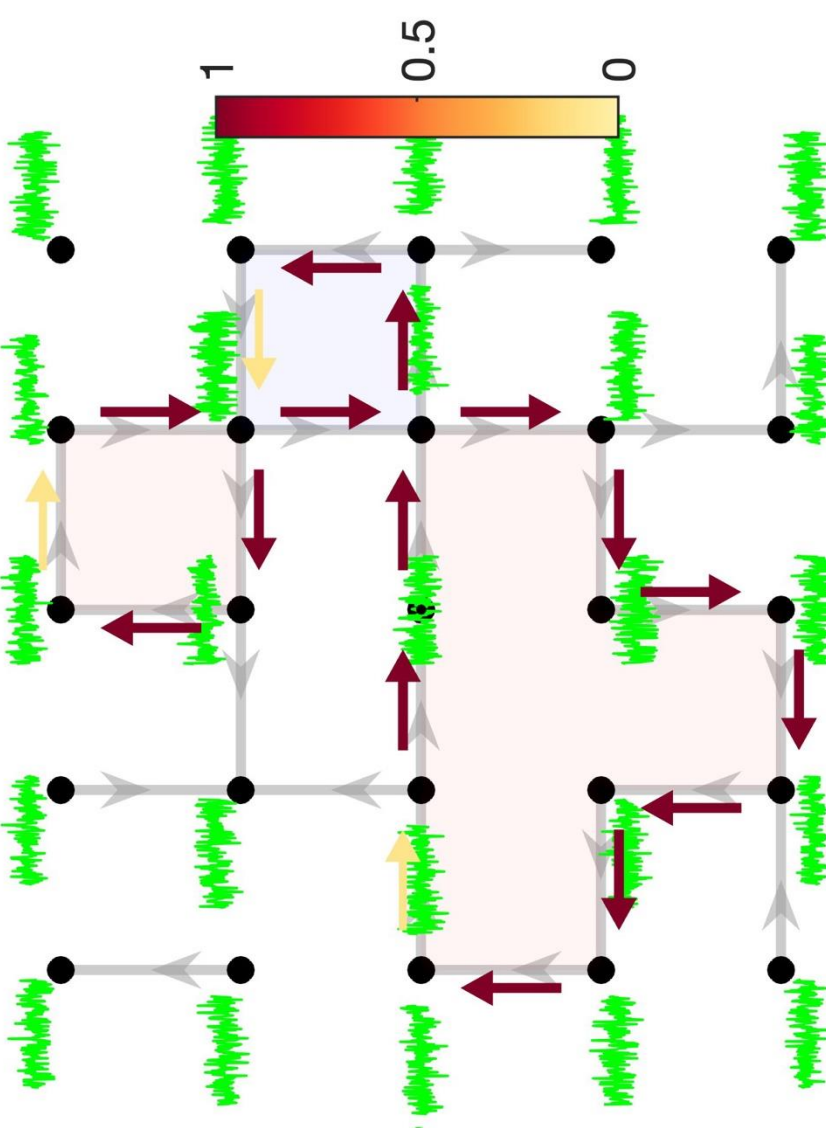
Rayleigh test

$$\beta_1 n \|\bar{U}\|_2^2 \xrightarrow{d} \chi_{\beta_1}^2$$

$$p\text{-value} = 10^{-29}$$

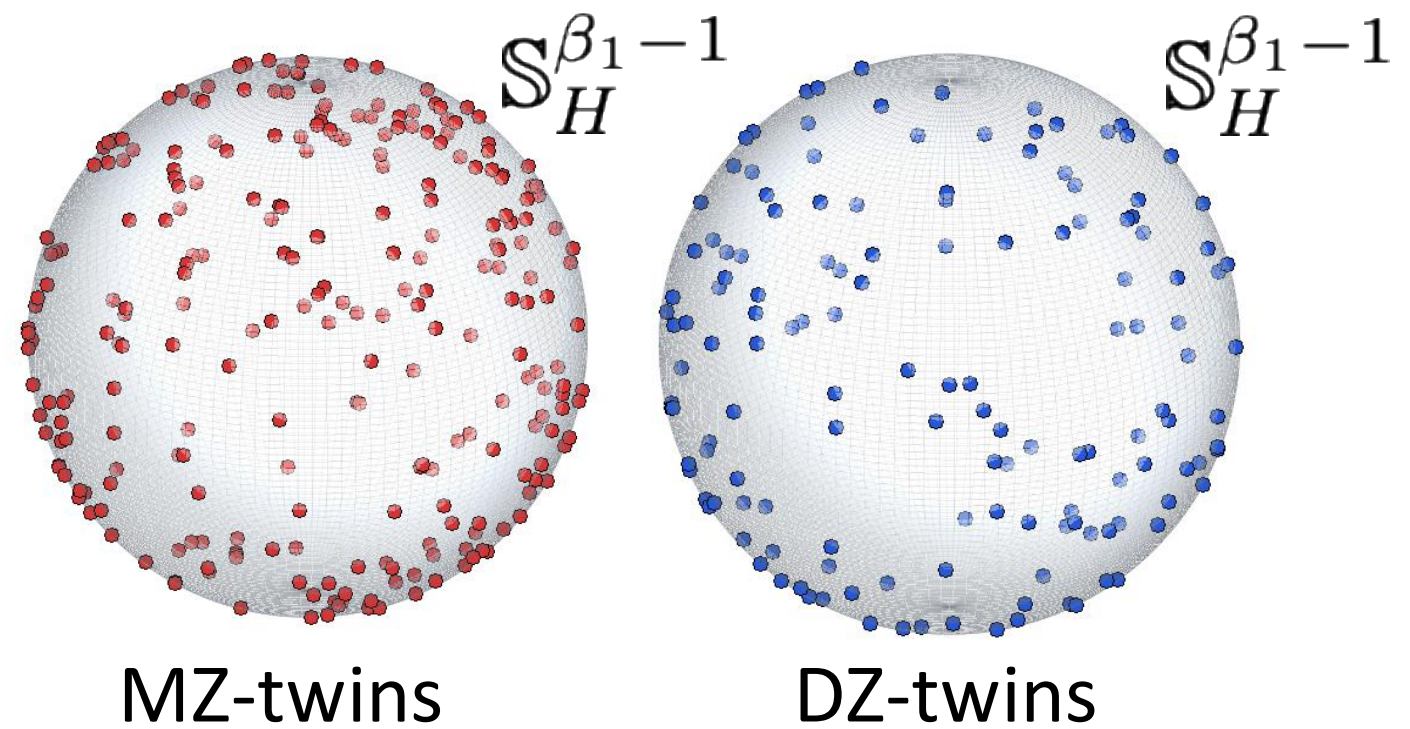
Performance – 3 broken cycle

Ground truth: 3 broken cycles



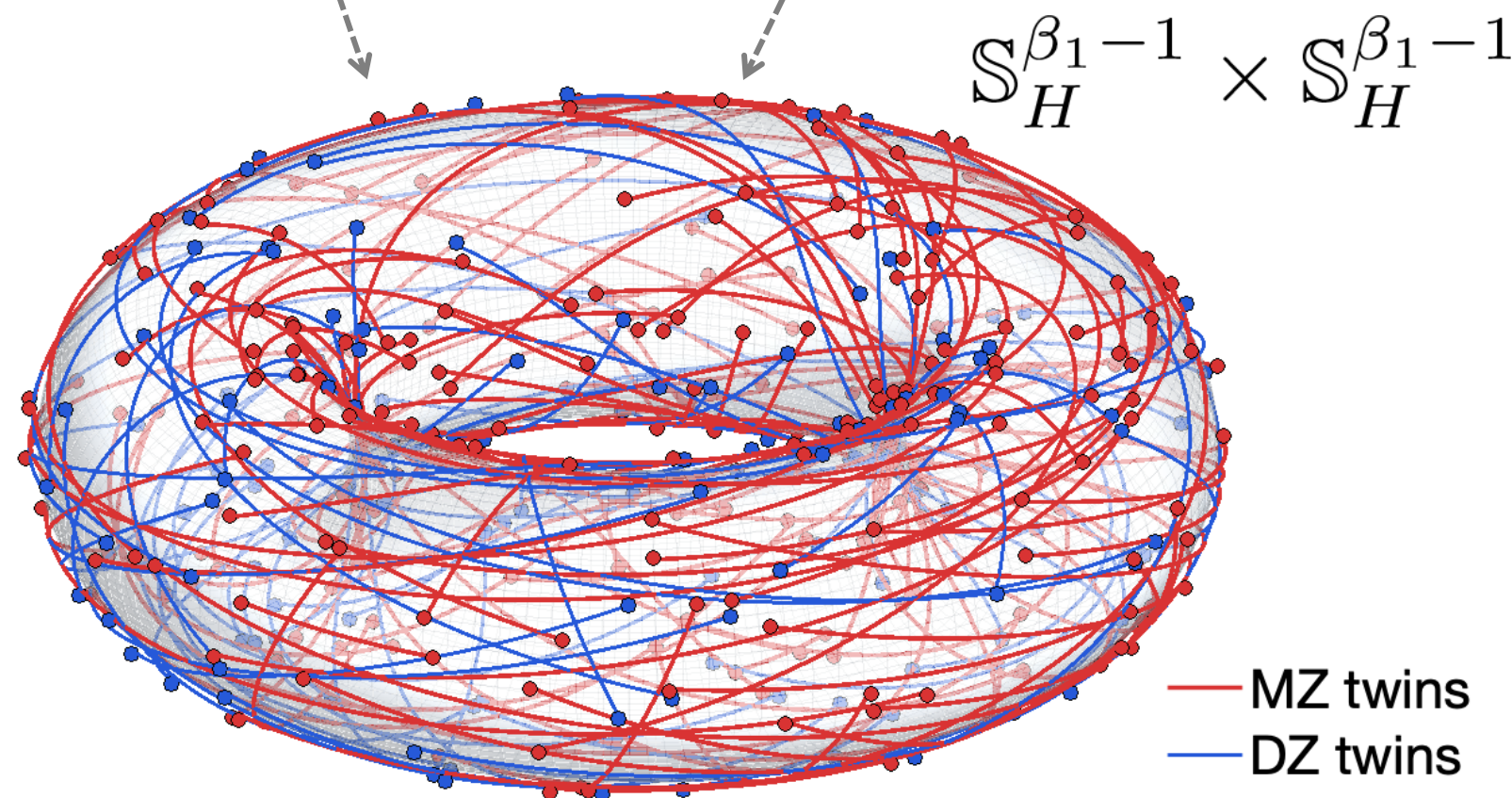
Heritability of cyclic interactions

Original directed
interactions
 p -value: 0.0923



Toroidal embedding

Geodesic distance
MZ-twin: 0.430π
DZ-twin: 0.488π
 p -value: 0.0305



Thank you. We are here.

